

Data Tables Specification

Complete Database Tables Documentation

Table 1: companies (Tenant Root)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique company identifier
name	VARCHAR	255	NO	-	-	Company legal name
registration_number	VARCHAR	100	NO	-	UK	Government registration number
email	VARCHAR	255	NO	-	-	Primary company email
phone	VARCHAR	20	YES	NULL	-	Company phone number
address	TEXT	-	YES	NULL	-	Company physical address
city	VARCHAR	100	YES	NULL	-	Company city
state	VARCHAR	100	YES	NULL	-	Company state/province
country	VARCHAR	100	YES	'USA'	-	Company country
postal_code	VARCHAR	20	YES	NULL	-	Postal/ZIP code
website	VARCHAR	255	YES	NULL	-	Company website URL
industry	VARCHAR	100	YES	NULL	IDX	Industry classification
company_size	ENUM	-	YES	'51-200'	IDX	Employee count range
logo_path	VARCHAR	255	YES	NULL	-	Local logo file path
logo_url	VARCHAR	500	YES	NULL	-	External logo URL
status	ENUM	-	YES	'active'	IDX	Company status
subscription_plan	ENUM	-	YES	'professional'	-	Subscription tier
subscription_expires	DATE	-	YES	NULL	-	Subscription expiry date
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Enums:

- company_size: '1-10', '11-50', '51-200', '201-500', '501-1000', '1000+'
- status: 'active', 'inactive', 'suspended'
- subscription_plan: 'free', 'basic', 'professional', 'enterprise'

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_companies_registration (registration_number)
- INDEX idx_companies_status (status)
- INDEX idx_companies_industry (industry)

Table 2: roles (System Reference)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique role identifier
name	VARCHAR	50	NO	-	UK	Role name
description	VARCHAR	255	YES	NULL	-	Role description
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time

Default Data:

- Admin - Company administrator with full access
- HR - Human Resources manager with employee management access
- Employee - Regular employee with self-service access

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_roles_name (name)

Table 3: permissions (System Reference)

Column	Data Type	Length	Null	Default	Key	Description
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id	INT	-	NO	AUTO_INCREMENT	PK	Unique permission identifier
name	VARCHAR	100	NO	-	UK	Permission name
module	VARCHAR	50	NO	-	IDX	Module/feature group
description	VARCHAR	255	YES	NULL	-	Permission description
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time

Permission Modules:

- employee (6 permissions)
- attendance (6 permissions)
- leave (6 permissions)
- payroll (5 permissions)
- company (2 permissions)
- user (4 permissions)

Total Permissions: 29 granular permissions

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_permissions_name (name)
- INDEX idx_permissions_module (module)

Table 4: role_permissions (Junction Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique mapping identifier
role_id	INT	-	NO	-	FK	Reference to roles table
permission_id	INT	-	NO	-	FK	Reference to permissions table

Relationships:

- Admin Role: All 29 permissions
- HR Role: 20 permissions (no delete operations)
- Employee Role: 6 self-service permissions

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_role_permission (role_id, permission_id)
- FOREIGN KEY fk_rp_role (role_id) REFERENCES roles(id)
- FOREIGN KEY fk_rp_permission (permission_id) REFERENCES permissions(id)

Table 5: users (Tenant Entity)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique user identifier
company_id	INT	-	NO	-	FK, IDX	Company association
role_id	INT	-	NO	-	FK	User role assignment
email	VARCHAR	255	NO	-	UK	User email address
password_hash	VARCHAR	255	NO	-	-	Encrypted password
status	ENUM	-	YES	'active'	IDX	User account status
last_login	TIMESTAMP	-	YES	NULL	-	Last login timestamp
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Enums:

- status: 'active', 'inactive', 'locked'

Security Features:

- bcrypt password hashing (cost factor 12)
- Account lockout after 5 failed attempts
- Session-based authentication
- Email uniqueness across system

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_users_email (email)
- INDEX idx_users_company (company_id)
- INDEX idx_users_status (status)
- FOREIGN KEY fk_users_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_users_role (role_id) REFERENCES roles(id)

Table 6: employees (Core Entity)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique employee identifier
company_id	INT	-	NO	-	FK, IDX	Company association
user_id	INT	-	YES	NULL	FK, UK	Optional user account
employee_code	VARCHAR	50	NO	-	-	Company-specific employee ID
first_name	VARCHAR	100	NO	-	-	Employee first name
last_name	VARCHAR	100	NO	-	-	Employee last name
email	VARCHAR	255	NO	-	-	Employee email address
phone	VARCHAR	20	YES	NULL	-	Employee phone number
date_of_birth	DATE	-	YES	NULL	-	Employee birth date
gender	ENUM	-	YES	NULL	-	Employee gender
address	TEXT	-	YES	NULL	-	Employee address
hire_date	DATE	-	NO	-	-	Employment start date
termination_date	DATE	-	YES	NULL	-	Employment end date
department	VARCHAR	100	YES	NULL	IDX	Employee department
designation	VARCHAR	100	YES	NULL	-	Job title/position
employment_type	ENUM	-	YES	'full_time'	-	Employment classification
status	ENUM	-	YES	'active'	IDX	Employee status
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Enums:

- gender: 'male', 'female', 'other'
- employment_type: 'full_time', 'part_time', 'contract', 'intern'
- status: 'active', 'inactive', 'terminated'

Business Rules:

- Employee code must be unique within company
- User account is optional (employees can exist without login)
- One user account can have only one employee profile

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_employees_user (user_id)
- UNIQUE KEY uk_employees_code (company_id, employee_code)
- INDEX idx_employees_company (company_id)
- INDEX idx_employees_status (status)
- INDEX idx_employees_department (department)
- FOREIGN KEY fk_employees_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_employees_user (user_id) REFERENCES users(id)

Table 7: attendance (Transaction Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique attendance identifier
company_id	INT	-	NO	-	FK, IDX	Company association
employee_id	INT	-	NO	-	FK	Employee reference
attendance_date	DATE	-	NO	-	IDX	Attendance date
clock_in_time	TIME	-	YES	NULL	-	Clock in time
clock_out_time	TIME	-	YES	NULL	-	Clock out time
total_hours	DECIMAL	4,2	YES	NULL	-	Total hours worked
status	ENUM	-	YES	'present'	IDX	Attendance status

notes	TEXT	-	YES NULL	-	Additional notes
created_at	TIMESTAMP	-	YES CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES CURRENT_TIMESTAMP	-	Last update time

Enums:

- status: 'present', 'absent', 'half_day', 'late', 'on_leave'

Business Rules:

- One attendance record per employee per day
- Total hours calculated automatically from clock times
- Status determines payroll calculations

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_attendance_employee_date (employee_id, attendance_date)
- INDEX idx_attendance_company (company_id)
- INDEX idx_attendance_date (attendance_date)
- INDEX idx_attendance_status (status)
- FOREIGN KEY fk_attendance_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_attendance_employee (employee_id) REFERENCES employees(id)

Table 8: leave_types (Configuration Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique leave type identifier
company_id	INT	-	NO	-	FK, IDX	Company association
name	VARCHAR	100	NO	-	-	Leave type name
annual_allocation	INT	-	NO	0	-	Days allocated per year
is_paid	TINYINT	1	YES	1	-	Paid leave flag
is_active	TINYINT	1	YES	1	-	Active status flag
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Default Leave Types (Created on Company Registration):

- Annual Leave: 20 days/year (Paid)
- Sick Leave: 10 days/year (Paid)
- Personal Leave: 5 days/year (Paid)
- Unpaid Leave: Unlimited (Unpaid)

Business Rules:

- Leave type names must be unique within company
- Annual allocation defines yearly entitlement
- Paid flag affects payroll calculations

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_leave_types_name (company_id, name)
- INDEX idx_leave_types_company (company_id)
- FOREIGN KEY fk_leave_types_company (company_id) REFERENCES companies(id)

Table 9: leave_requests (Transaction Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique request identifier
company_id	INT	-	NO	-	FK, IDX	Company association
employee_id	INT	-	NO	-	FK, IDX	Employee making request
leave_type_id	INT	-	NO	-	FK, IDX	Type of leave requested
start_date	DATE	-	NO	-	IDX	Leave start date
end_date	DATE	-	NO	-	IDX	Leave end date
total_days	INT	-	NO	-	-	Total days requested
reason	TEXT	-	YES NULL	-	-	Reason for leave

status	ENUM	-	YES 'pending'	IDX	Request status
approver_id	INT	-	YES NULL	FK	User who approved/rejected
approval_date	TIMESTAMP	-	YES NULL	-	Approval/rejection date
rejection_reason	TEXT	-	YES NULL	-	Reason for rejection
created_at	TIMESTAMP	-	YES CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES CURRENT_TIMESTAMP	-	Last update time

Enums:

- status: 'pending', 'approved', 'rejected', 'cancelled'

Workflow:

1. Employee submits request (status: pending)
2. Manager/HR reviews request
3. Request approved/rejected with optional reason
4. Employee notified of decision

Indexes:

- PRIMARY KEY (id)
- INDEX idx_leave_requests_company (company_id)
- INDEX idx_leave_requests_employee (employee_id)
- INDEX idx_leave_requests_type (leave_type_id)
- INDEX idx_leave_requests_status (status)
- INDEX idx_leave_requests_dates (start_date, end_date)
- FOREIGN KEY fk_leave_requests_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_leave_requests_employee (employee_id) REFERENCES employees(id)
- FOREIGN KEY fk_leave_requests_type (leave_type_id) REFERENCES leave_types(id)
- FOREIGN KEY fk_leave_requests_approver (approver_id) REFERENCES users(id)

Table 10: salary_structures (Configuration Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique structure identifier
company_id	INT	-	NO	-	FK, IDX	Company association
employee_id	INT	-	NO	-	FK, IDX	Employee reference
basic_salary	DECIMAL	12,2	NO	-	-	Base salary amount
housing_allowance	DECIMAL	12,2	YES	0.00	-	Housing allowance
transport_allowance	DECIMAL	12,2	YES	0.00	-	Transport allowance
other_allowances	DECIMAL	12,2	YES	0.00	-	Other allowances
tax_deduction	DECIMAL	12,2	YES	0.00	-	Tax deduction amount
insurance_deduction	DECIMAL	12,2	YES	0.00	-	Insurance deduction
other_deductions	DECIMAL	12,2	YES	0.00	-	Other deductions
effective_date	DATE	-	NO	-	IDX	Structure effective date
is_current	TINYINT	1	YES	1	IDX	Current structure flag
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Currency: Indian Rupee (₹ INR)

Salary Components:

- **Gross Salary** = basic_salary + housing_allowance + transport_allowance + other_allowances
- **Total Deductions** = tax_deduction + insurance_deduction + other_deductions
- **Net Salary** = Gross Salary - Total Deductions

Business Rules:

- Only one current salary structure per employee
- Historical structures maintained for audit trail
- Effective date determines when structure becomes active

Indexes:

- PRIMARY KEY (id)
- INDEX idx_salary_company (company_id)
- INDEX idx_salary_employee (employee_id)

- INDEX idx_salary_effective (effective_date)
- INDEX idx_salary_current (is_current)
- FOREIGN KEY fk_salary_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_salary_employee (employee_id) REFERENCES employees(id)

Table 11: payroll_records (Transaction Table)

Column	Data Type	Length	Null	Default	Key	Description
id	INT	-	NO	AUTO_INCREMENT	PK	Unique payroll identifier
company_id	INT	-	NO	-	FK, IDX	Company association
employee_id	INT	-	NO	-	FK, IDX	Employee reference
salary_structure_id	INT	-	NO	-	FK	Salary structure used
year	INT	-	NO	-	IDX	Payroll year
month	INT	-	NO	-	IDX	Payroll month (1-12)
gross_salary	DECIMAL	12,2	NO	-	-	Total gross amount
total_deductions	DECIMAL	12,2	NO	-	-	Total deduction amount
net_salary	DECIMAL	12,2	NO	-	-	Final payable amount
payment_date	DATE	-	YES	NULL	-	Actual payment date
payment_status	ENUM	-	YES	'pending'	IDX	Payment status
payment_reference	VARCHAR	100	YES	NULL	-	Payment reference number
created_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Record creation time
updated_at	TIMESTAMP	-	YES	CURRENT_TIMESTAMP	-	Last update time

Enums:

- payment_status: 'pending', 'processed', 'paid', 'failed'

Business Rules:

- One payroll record per employee per month
- Month must be between 1 and 12 (check constraint)
- Immutable once payment is processed
- Links to salary structure for audit trail

Payment Workflow :

1. Payroll processing creates records (status: pending)
2. Calculations verified and approved (status: processed)
3. Payment executed (status: paid)
4. Failed payments marked (status: failed)

Indexes:

- PRIMARY KEY (id)
- UNIQUE KEY uk_payroll_employee_month (employee_id, year, month)
- INDEX idx_payroll_company (company_id)
- INDEX idx_payroll_employee (employee_id)
- INDEX idx_payroll_period (year, month)
- INDEX idx_payroll_status (payment_status)
- FOREIGN KEY fk_payroll_company (company_id) REFERENCES companies(id)
- FOREIGN KEY fk_payroll_employee (employee_id) REFERENCES employees(id)
- FOREIGN KEY fk_payroll_salary (salary_structure_id) REFERENCES salary_structures(id)
- CHECK CONSTRAINT chk_payroll_month (month >= 1 AND month <= 12)

Database Summary Statistics

Table Count and Types

- **Total Tables:** 11
- **Reference Tables:** 3 (roles, permissions, role_permissions)
- **Tenant Root:** 1 (companies)
- **Tenant Tables:** 7 (users, employees, attendance, leave_types, leave_requests, salary_structures, payroll_records)

Constraint Summary

- **Primary Keys:** 11
- **Foreign Keys:** 15

- **Unique Constraints:** 8
- **Check Constraints:** 1
- **Indexes:** 35+

Data Type Usage

- **INT:** 35 columns (identifiers, counts, years)
- **VARCHAR:** 25 columns (names, codes, references)
- **DECIMAL:** 10 columns (financial amounts)
- **DATE:** 8 columns (dates)
- **TIME:** 2 columns (clock times)
- **TIMESTAMP:** 22 columns (audit trail)
- **TEXT:** 5 columns (long descriptions)
- **ENUM:** 8 columns (controlled vocabularies)
- **TINYINT:** 4 columns (boolean flags)

This comprehensive data tables specification provides complete documentation of all database structures, constraints, and business rules for the multi-tenant HRMS system.